CSE1030
C# PROGRAMMING AND .NETFARMEWORK

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INDEX

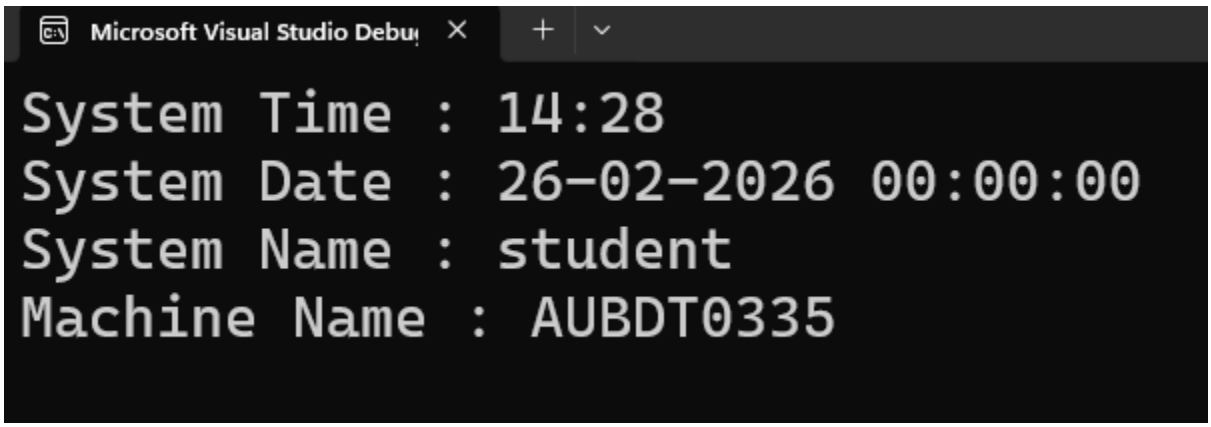
Sl.No.	Program Title	Page No.	Signature
1.	Write a C# program to display system date , time and user details		
2.	Write a program to perform arithmetic , relational and logical operations using user input		
3.	Write a program to check eligibility for voting using conditional statements.		
4.	Program to generate multiplication table using loops		
5.	Write a program to generate Fibonacci series and amstrong numbers.		
6.	Write a C# program to find fractorial of a number and the sum of digits using Recursion		
7.	Write a C# program to validate a username and password using conditional logic.		
8.	Write a C# program to demonstrate one dimensional array operations such as		

	calculating the sum and average of elements.		
9.	Write a C# program to sort elements in an array in ascending order.		
10.	Write a C# program to demonstrate string operations (length, concatenation, comparison).		
11.	Write the sample program to print a different greeting, for example, "Good Day!"		
12.	Write a console application that prints your first and last name on the console.		
13.	Write a program that prints the following numbers on the console: 1, 101, 1001, each on a new line.		
14.	Write a program that prints on the console the current date and time.		
15.	Write a program that prints the square root of 12345.		
16.	Write a program that prints the first 100 members of the sequence 2, -3, 4, -5, 6, -7, 8.		
17.	Write a program that reads your age from the console and prints your age after 10 years.		

01. Write a C# program to display system date , time and user details

```
using System;
using System.Data;
using System.Threading.Tasks;

namespace FirstProgram
{
   internal class Program
   {
      static void Main(string[] args)
      {
         Console.WriteLine("System Time : " + DateTime.Now.ToShortTimeString());
         Console.WriteLine("System Date : " + DateTime.Today.ToString());
         Console.WriteLine("System Name : " + Environment.UserName);
         Console.WriteLine("Machine Name : "+ Environment.MachineName);
      }
}
}
```

OUTPUT



02. Write a program to perform arithmetic , relational and logical operations using user input

```
using System;
class Program
{
   static void Main()
   {

      Console.Write("Enter first number: ");
      int num1 = Convert.ToInt32(Console.ReadLine());

      Console.Write("Enter second number: ");
      int num2 = Convert.ToInt32(Console.ReadLine());
```

```
        Console.WriteLine("--- Arithmetic Operations ---");
        Console.WriteLine("Addition: " + (num1 + num2));
        Console.WriteLine("Subtraction: " + (num1 - num2));
        Console.WriteLine("Multiplication: " + (num1 * num2));
        Console.WriteLine("Division: " + (num1 / num2));
        Console.WriteLine("Modulus: " + (num1 % num2));

        Console.WriteLine("--- Relational Operations ---");
        Console.WriteLine("Equal: " + (num1 == num2));
        Console.WriteLine("Not Equal: " + (num1 != num2));
        Console.WriteLine("Greater Than: " + (num1 > num2));
        Console.WriteLine("Less Than: " + (num1 < num2));
        Console.WriteLine("Greater or Equal: " + (num1 >= num2));
        Console.WriteLine("Less or Equal: " + (num1 <= num2));

        Console.WriteLine("--- Logical Operations ---");
        Console.WriteLine("AND (num1 > 0 && num2 > 0): " + (num1 > 0 && num2 > 0));
        Console.WriteLine("OR (num1 > 0 || num2 > 0): " + (num1 > 0 || num2 > 0));
        Console.WriteLine("NOT (num1 > 0): " + !(num1 > 0));
        Console.ReadLine();
    }
}
```

OUTPUT

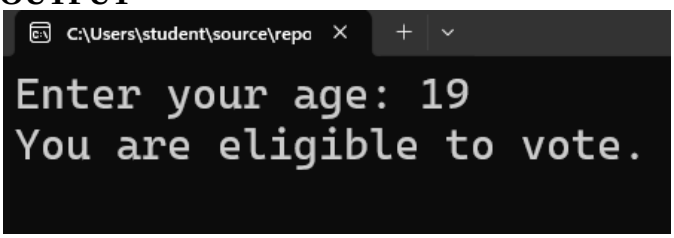
```
C:\Users\student\source\repo
Enter first number: 12
Enter second number: 15
--- Arithmetic Operations ---
Addition: 27
Subtraction: -3
Multiplication: 180
Division: 0
Modulus: 12
--- Relational Operations ---
Equal: False
Not Equal: True
Greater Than: False
Less Than: True
Greater or Equal: False
Less or Equal: True
--- Logical Operations ---
AND (num1 > 0 && num2 > 0): True
OR (num1 > 0 || num2 > 0): True
NOT (num1 > 0): False
```

03. Write a program to check eligibility for voting using conditional statements.

```
using System;
using System.Collections.Generic;
using System.Linq;
```

```
using System.Text;
using System.Threading.Tasks;

namespace _3_program
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter your age: ");
            int age = Convert.ToInt32(Console.ReadLine());

            if (age >= 18)
            {
                Console.WriteLine("You are eligible to vote.");
            }
            else
            {
                Console.WriteLine("You are not eligible to vote.");
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT

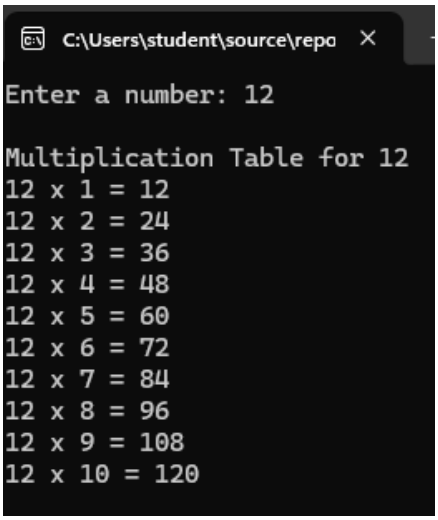


04. Program to generate multiplication table using loops

```
using System;
class Program
{
    static void Main()
    {
        Console.Write("Enter a number: ");
        int number = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("\nMultiplication Table for " + number);

        for (int i = 1; i <= 10; i++)
        {
            Console.WriteLine(number + " x " + i + " = " + (number * i));
        }
```

```
        Console.ReadLine();
    }
}
```

OUTPUT



05. Write a program to generate Fibonacci series and amstrong numbers

```
using System;
class Program
{
    static void Main()
    {
        Console.Write("Enter the number of terms for the Fibonacci series: ");
        int n = Convert.ToInt32(Console.ReadLine());

        int first = 0, second = 1;

        Console.WriteLine("\nFibonacci Series:");
        for (int i = 1; i <= n; i++)
        {
            Console.Write(first + " ");
            int next = first + second;
            first = second;
            second = next;
        }

        Console.WriteLine();
        Console.ReadLine();
    }
}
```

OUTPUT

```
C:\Users\student\source\repo
Enter the number of terms for the Fibonacci series: 5

Fibonacci Series:
0 1 1 2 3
```

06. Program to find factorial and sum of digits using recursion

```
using System;

class RecursionExample
{
    static int Factorial(int n)
    {
        if (n == 0 || n == 1)
            return 1;
        return n * Factorial(n - 1);
    }

    static int SumOfDigits(int n)
    {
        if (n == 0)
            return 0;
        return (n % 10) + SumOfDigits(n / 10);
    }

    static void Main()
    {
        Console.Write("Enter a number: ");
        int num = int.Parse(Console.ReadLine());

        Console.WriteLine("Factorial: " + Factorial(num));
        Console.WriteLine("Sum of Digits: " + SumOfDigits(num));
    }
```

OUTPUT

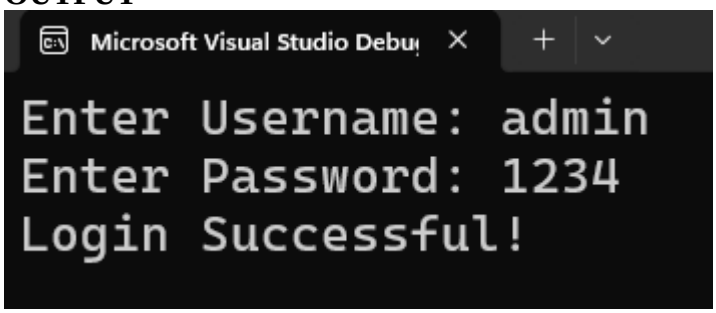
```
Microsoft Visual Studio Debu
Enter a number: 21
Factorial: -1195114496
Sum of Digits: 3

C:\Users\student\source\rep
To automatically close the
 close the console when deb
Press any key to close this
```

07. Program to validate a username and password using conditional logic.

```
using System;
```

```
class LoginValidation
{
   static void Main()
   {
      string correctUsername = "admin";
      string correctPassword = "1234";

      Console.Write("Enter Username: ");
      string username = Console.ReadLine();

      Console.Write("Enter Password: ");
      string password = Console.ReadLine();

      if (username == correctUsername && password == correctPassword)
      {
         Console.WriteLine("Login Successful!");
      }
      else
      {
         Console.WriteLine("Invalid Username or Password.");
      }
   }
}
```

OUTPUT

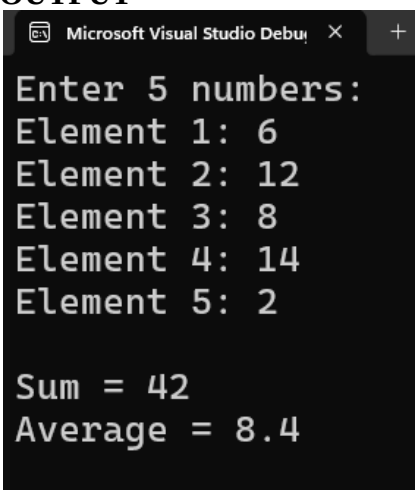


08. Program to demonstrate one-dimensional array operations (sum, average)

```
using System;
```

```
class ArrayOperations
{
    static void Main()
    {
        int[] numbers = new int[5];
        int sum = 0;

        Console.WriteLine("Enter 5 numbers:");
        // Input elements into array
        for (int i = 0; i < numbers.Length; i++)
        {
            Console.Write("Element " + (i + 1) + ": ");
            numbers[i] = int.Parse(Console.ReadLine());
        }
        // Calculate sum
        for (int i = 0; i < numbers.Length; i++)
        {
            sum += numbers[i];
        }
        // Calculate average
        double average = (double)sum / numbers.Length;
        Console.WriteLine("\nSum = " + sum);
        Console.WriteLine("Average = " + average);
    }
}
```

OUTPUT

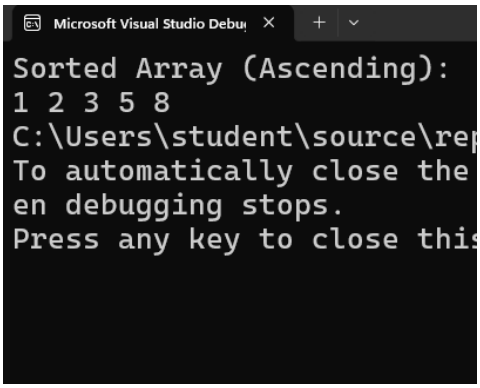


09. Program to sort elements in an array in ascending order.

```
using System;
```

```
class SortArray
{
    static void Main()
    {
        int[] arr = { 5, 2, 8, 1, 3 };

        Array.Sort(arr);

        Console.WriteLine("Sorted Array (Ascending):");
        foreach (int num in arr)
        {
            Console.Write(num + " ");
        }
    }
}
```

OUTPUT

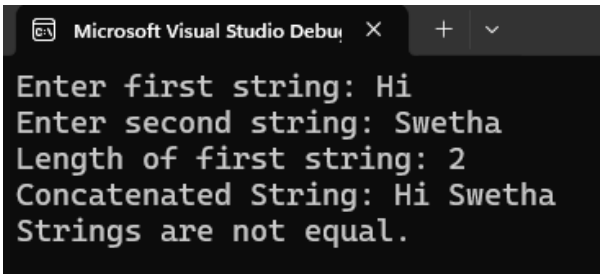


10. Program to demonstrate string operations (length, concatenation, comparison).

```
using System;

class StringOperations
{
    static void Main()
    {
        Console.Write("Enter first string: ");
        string str1 = Console.ReadLine();

        Console.Write("Enter second string: ");
        string str2 = Console.ReadLine();

        // Length
        Console.WriteLine("Length of first string: " + str1.Length);

        // Concatenation
        string combined = str1 + " " + str2;
        Console.WriteLine("Concatenated String: " + combined);
```

```
        // Comparison
        if (str1.Equals(str2))
        {
            Console.WriteLine("Both strings are equal.");
        }
        else
        {
            Console.WriteLine("Strings are not equal.");
        }
    }
}
```

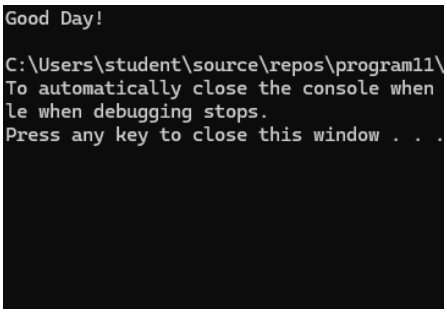
OUTPUT



11. Write the sample program to print a different greeting, for example, "Good Day!"

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Good Day!");
    }
}
```

OUTPUT

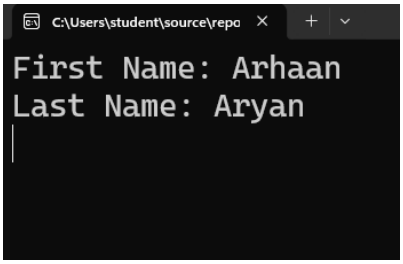


12. Write a console application that prints your first and last name on the console.

```
using System;
```

```
class Program
{
   static void Main(string[] args)
   {
      string firstName = "Arhaan";
      string lastName = "Aryan";
      Console.WriteLine("First Name: " + firstName);
      Console.WriteLine("Last Name: " + lastName);
      Console.ReadKey();
   }
}
```

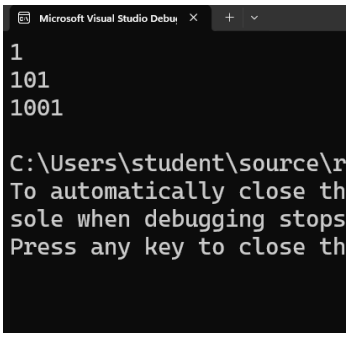
OUTPUT



13. Write a program that prints the following numbers on the console: 1, 101, 1001, each on a new line.

```
using System;

class Program
{
   static void Main()
   {
      Console.WriteLine(1);
      Console.WriteLine(101);
      Console.WriteLine(1001);
   }
}
```

OUTPUT



14. Write a program that prints on the console the current date and time.

```
using System;
```

```
class Program
{
   static void Main()
   {
      DateTime now = DateTime.Now;
      Console.WriteLine("Current Date and Time: " + now);
   }
}
```

OUTPUT

```
Microsoft Visual Studio Debu
Current Date and Time: 12-03-2026 13:53:47

C:\Users\student\source\repos\program14\bin\D
```

15. Write a program that prints the square root of 12345.

```
using System;

class Program
{
   static void Main()
   {
      double number = 12345;
      double result = Math.Sqrt(number);
      Console.WriteLine("The square root of " + number + " is " + result);
   }
}
```

OUTPUT

```
Microsoft Visual Studio Debu
The square root of 12345 is 111.108055513541

C:\Users\student\source\repos\program15\bin\De
.
To automatically close the console when debugg
 close the console when debugging stops.
Press any key to close this window . . .
```

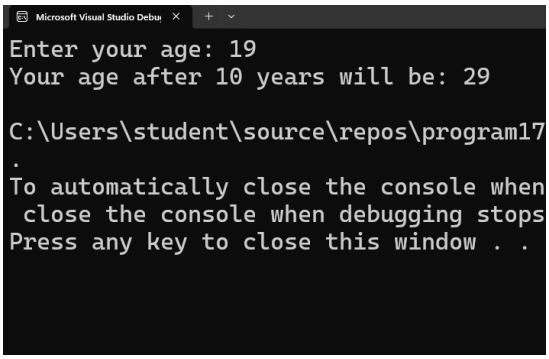
16. Write a program that prints the first 100 members of the sequence 2, -3, 4, -5, 6, -7, 8.

```
using System;

class Program
{
    static void Main()
    {
        int count = 100;
        for (int i = 0; i < count; i++)
        {
            int number = i + 2;
            if (i % 2 == 1)
            {
                number = -number;
            }
            Console.Write(number);
            if (i < count - 1)
            {
                Console.Write(", ");
            }
        }
    }
}
```

OUTPUT

```
2, -3, 4, -5, 6, -7, 8, -9, 10, -11, 12, -13, 14, -15, 16, -17, 18, -19, 20, -21, 22, -23, 24, -25, 26,
-27, 28, -29, 30, -31, 32, -33, 34, -35, 36, -37, 38, -39, 40, -41, 42, -43, 44, -45, 46, -47, 48, -49,
50, -51, 52, -53, 54, -55, 56, -57, 58, -59, 60, -61, 62, -63, 64, -65, 66, -67, 68, -69, 70, -71, 72, -
73, 74, -75, 76, -77, 78, -79, 80, -81, 82, -83, 84, -85, 86, -87, 88, -89, 90, -91, 92, -93, 94, -95, 9
6, -97, 98, -99, 100, -101
```

17. Write a program that reads your age from the console and prints yourage after 10 years.

```
using System;

class Program
{
   static void Main()
   {
      Console.Write("Enter your age: ");
      int age = Convert.ToInt32(Console.ReadLine());
      int futureAge = age + 10;
      Console.WriteLine("Your age after 10 years will be: " + futureAge);
   }
}
```

OUTPUT